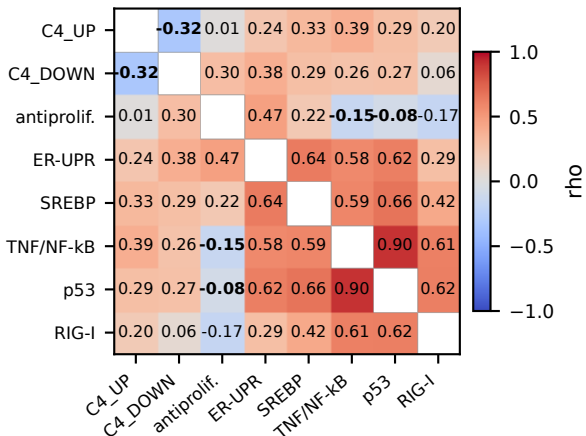
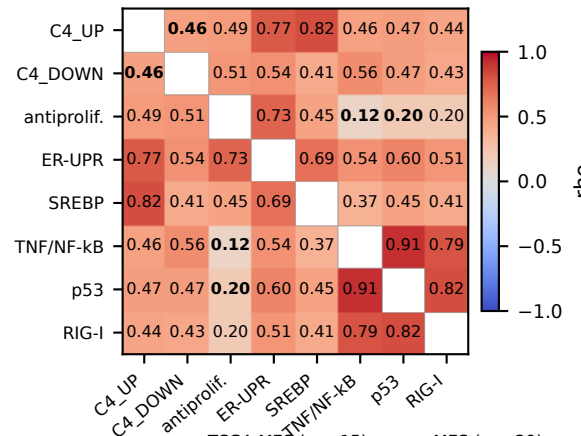


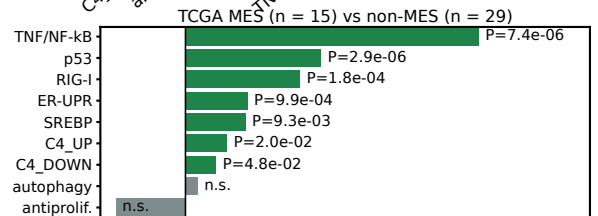
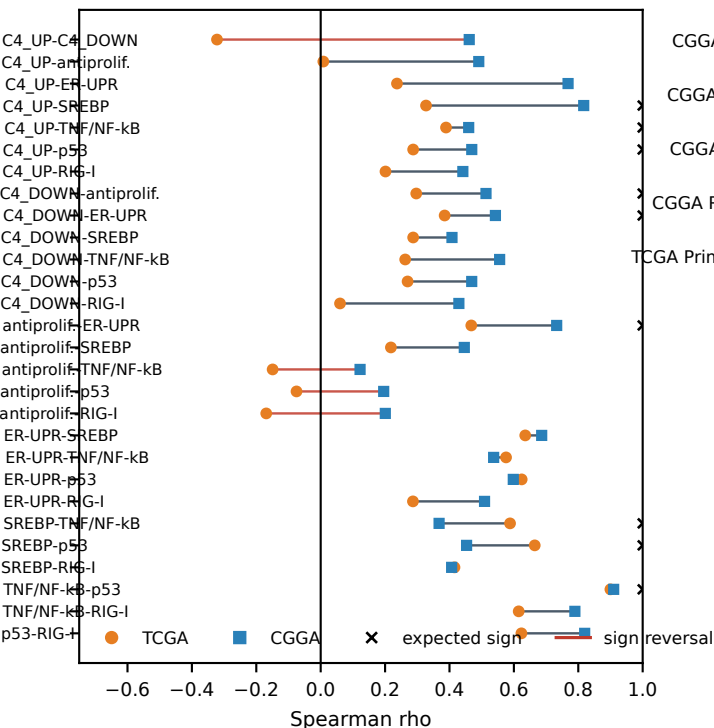
A (A) TCGA relationship structure (n = 288); Spearman rho between module scores



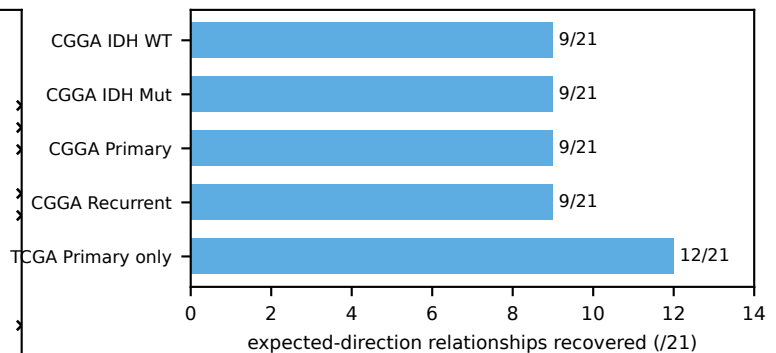
B (B) CGGA relationship structure (n = 249); Spearman rho between module scores



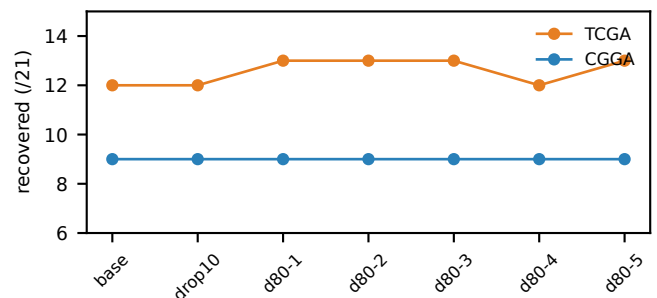
C Cross-cohort forest plot of the 28 module relationships (circles: TCGA; squares: CGGA; x: expected sign; red: 3 sign-reversed pairs)



D Robustness and negative results: stratification (IDH, primary/recurrent)



gene-perturbation sensitivity (no imputation); coverage rule >= 0.80



Panel A/B note: Contextualisation only: these are cross-sectional correlations in bulk tumours, not causal or replication evidence.

Bold values mark the three sign-reversed pairs.

Panel C summary (robust_1_cross_cohort_sign): reversals within expected set: C4_UP-C4_DOWN (T -0.32/C +0.46),

M_shared_antiproliferative-M_TNF_NFkB (T -0.15/C +0.12), M_shared_antiproliferative-M_p53_apoptosis (T -0.08/C +0.20); RIG-I pair no expected sign (T -0.17/C +0.20)

Panel D values (frozen C3_ROBUSTNESS.tsv): stratification dir_match CGGA_IDH_WT=9/21; CGGA_IDH_Mut=9/21; CGGA_Primary=9/21; CGGA_Recurrent=9/21; TCGA_Primary_only=12/21; coverage TCGA min=0.970 (C5_UP); CGGA min=0.865 (C5_DOWN).